

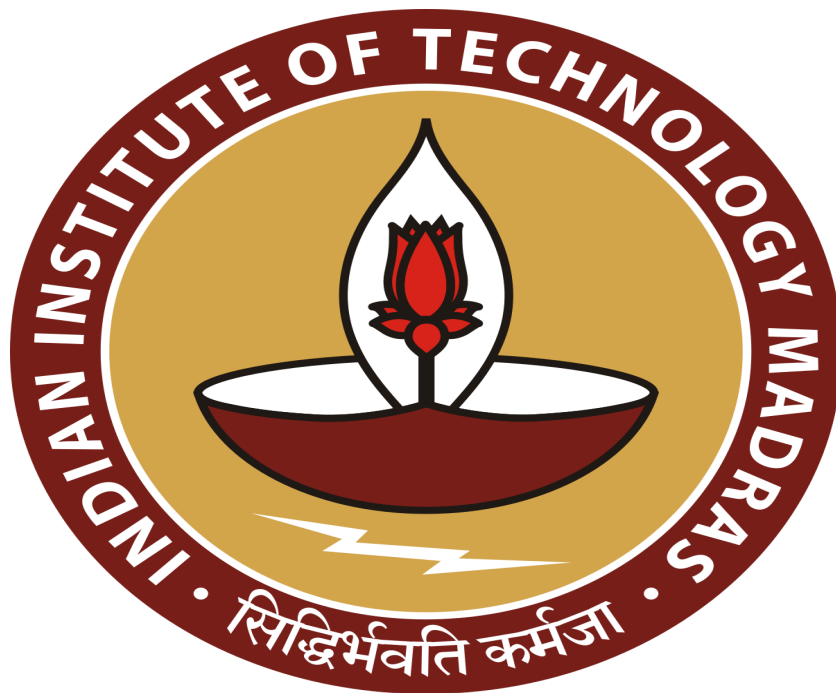
Enhancing Efficiency in Retail Pharmacies: Challenges and Solutions

A Proposal report for the BDM capstone Project

Submitted by

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Declaration Statement

I am working on a Project titled “**Enhancing Efficiency in Retail Pharmacies: Challenges and Solutions**”. I appreciate **SHRI BALAJI MEDICAL STORE** for providing the necessary resources that enabled me to conduct my project. I hereby assert that the data presented and assessed in this project report are genuine and precise to the utmost extent of my knowledge and capabilities. The data has been gathered through primary sources and carefully analyzed to ensure reliability. Additionally, I affirm that all procedures employed for data collection and analysis have been duly explained in this report. The outcomes and inferences derived from the data are an accurate depiction of the findings acquired through thorough analytical procedures. I am dedicated to adhering to the principles of academic honesty and integrity, and I am receptive to any additional examination or validation of the data contained in this project report. I understand that the execution of this project is intended for individual completion and is not to be undertaken collectively. I thus affirm that I am not engaged in any form of collaboration with other individuals, and that all the work undertaken has been solely conducted by me. If plagiarism is detected in the report at any stage of the project's completion, I am fully aware and prepared to accept disciplinary measures imposed by the relevant authority. I agree that all the recommendations are business-specific and limited to this project exclusively, and cannot be utilized for any other purpose with an IIT Madras tag. I understand that IIT Madras does not endorse this.



Signature of Candidate:

Name: Paras

Date: 13-05-2025

1. Executive Summary and Title

This project focuses on **Shri Balaji Medical Store**, a small retail pharmacy located in a rural village of **Haryana, India**. Operating under a **B2C model**, the store primarily sells **generic medicines, over-the-counter (OTC) drugs**, and a limited range of **veterinary medicines**. It is owned and managed by **Mr. Mangeram**, who oversees all operations manually without any data-driven inventory system.

The store faces several operational challenges, such as **frequent stockouts of essential generics, excess inventory of slow-moving veterinary items**, and **losses due to expired medicines**. Additionally, **low sales of higher-margin products** further reduce

overall profitability. These problems stem from a lack of structured inventory tracking and sales analysis.

The objective of this project is to address these issues by analyzing the store's transaction data using columns such as **medicine name, quantity, amount, and expiry date**. Through categorization and date-based analysis, the project will identify demand patterns, highlight near-expiry stock, and evaluate the profitability of various product types.

The expected outcome is to help the store owner make informed decisions about stock planning, prioritize high-margin items, avoid unnecessary overstocking, and reduce waste due to product expiry, ultimately improving profitability and operational efficiency.

2. Organization Background

Shri Balaji Medical Store is a small retail pharmacy located in a rural village in **Haryana, India**, owned by **Mr. Mangeram**. The store was established in **2021**, following the COVID-19 pandemic, with the vision of providing affordable and accessible healthcare to residents. Operating as a **B2C business**, the pharmacy offers **generic medicines, over-the-counter (OTC) drugs**, and a select range of **veterinary medicines** for farmers and pet owners.

Though it started as a small setup with limited medicines and customers, community trust helped the store grow quickly. With financial support from local well-wishers in 2022, the store expanded its inventory to include essential medicines like **Paracetamol, Albendazole**, and more. Mr. Mangeram is well known in the village and personally interacts with most customers. While the store is growing, it faces challenges in **inventory management, stock expiration**, and **regulatory compliance**, which the project aims to address using data analysis.

3. Problem Statement

Shri Balaji Medical Store is currently facing multiple challenges due to a lack of inventory tracking and data-driven decision-making. These include stock imbalances, poor sales of profitable items, and risks associated with expired products. This project aims to identify and solve these inefficiencies through transaction-level data analysis and insight generation.

3.1 Problem Statement 1:

The store frequently faces stockouts of high-demand medicines and overstocking of low-demand items, leading to missed sales and increased holding costs.

3.2 Problem Statement 2:

Most sales are from low-margin items, while high-margin medicines remain unsold due to a lack of targeted promotion and product visibility.

3.3 Problem Statement 3:

There is no system in place to track expiry dates, resulting in expired stock remaining on shelves, which can cause financial loss, regulatory penalties, and damage to customer trust.

4. Background of the Problem

Shri Balaji Medical Store, a small pharmacy located in rural Haryana, is currently facing multiple operational challenges that are negatively affecting profitability and customer satisfaction. The most critical issues include **poor inventory management**, **low sales of high-margin medicines**, and a **lack of monitoring for product expiry**, all of which lead to financial loss and regulatory risks.

These problems primarily arise due to the absence of structured data management. The store relies on **handwritten sales and inventory notes**, which are often **disorganized and rarely reviewed**. Without accurate tracking or analytical tools, the store fails to identify demand trends, manage stock levels efficiently, or make informed restocking decisions.

Internally, the lack of stock tracking systems, unstructured records, and limited training for staff contribute to poor operational efficiency. Externally, the store faces **growing competition from large pharmacies, online medicine platforms**, and **strict drug expiration regulations**. Additionally, **seasonal demand fluctuations** further complicate inventory planning.

The owner's limited visibility into sales performance and inventory turnover prevents strategic decision-making. This project aims to address these internal and external issues by applying business data management techniques to identify patterns, predict demand, and recommend actionable improvements.

5. Problem-Solving Approach

To address the operational challenges faced by **Shri Balaji Medical Store**, the project will follow a data-driven approach using structured analysis of the store's transaction data. The primary goal is to optimize inventory, improve sales of high-margin medicines, and reduce losses due to expired stock.

1. Sales and Inventory Pattern Analysis:

Using data columns such as “**Particulars,**” “**Quantity,**” “**Date,**” and “**Amount (Rs.)**”, we will analyze monthly sales volumes to determine which products are in high demand (e.g., **Paracetamol: 250 tablets/month**) and which are underperforming (e.g., **Oxytetracycline**). This will enable the store to maintain optimal stock levels—for instance, keeping **300 units** of Paracetamol while reducing veterinary stock to **20 units**. Additionally, seasonal trend analysis will help identify time-based demand patterns, such as higher antibiotic sales during winter.

2. Promotion of High-Margin Products:

By evaluating “**Amount (Rs.)**” to “**Particulars**” and “**Quantity**,” we will identify high-margin medicines such as **Betadine (Rs. 120)** or **Enrofloxacin**. Recommendations will be provided to promote these through product bundling, staff training (especially for OTC drugs tagged as “Doctor”), and upselling techniques to boost revenue.

3. Supplier Coordination:

Data from the “**Manufacturing Company**” column will be used to evaluate supplier performance. This will allow the store to negotiate better delivery timelines and return policies, especially for slow-moving or near-expiry stock.

4. Expiry Management:

To prevent losses from expired medicines and avoid regulatory penalties, we will monitor the “**Expiry Date,**” “**Batch No.,**” and “**Quantity**” fields. By tracking how quickly different batches sell, the store can prioritize the sale of older stock (e.g., selling **Paracetamol expiring in 2026** before the 2027 batch). If certain batches are lagging, discounting strategies or supplier return agreements can be implemented to manage the risk.

5. Customer Insight and Loyalty:

By analyzing the “**Purchaser/Patient**” data, the store can identify frequent customers and offer loyalty discounts or personalized deals, encouraging repeat visits and higher customer retention.

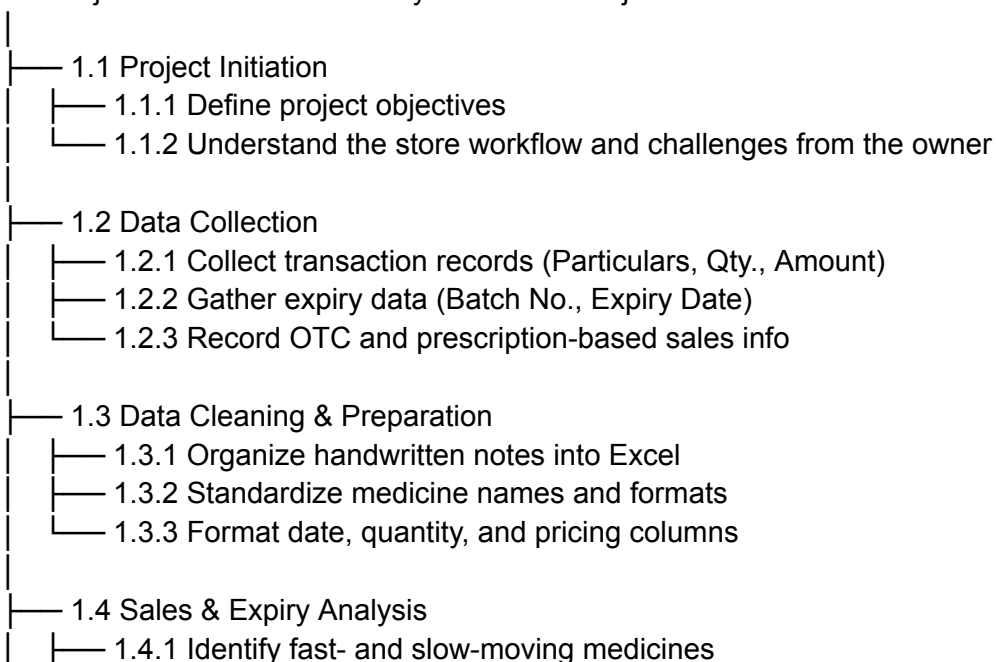
Overall, the project aims to apply Excel-based data tools to extract actionable insights that support better inventory control, increased profitability, and stronger customer satisfaction.

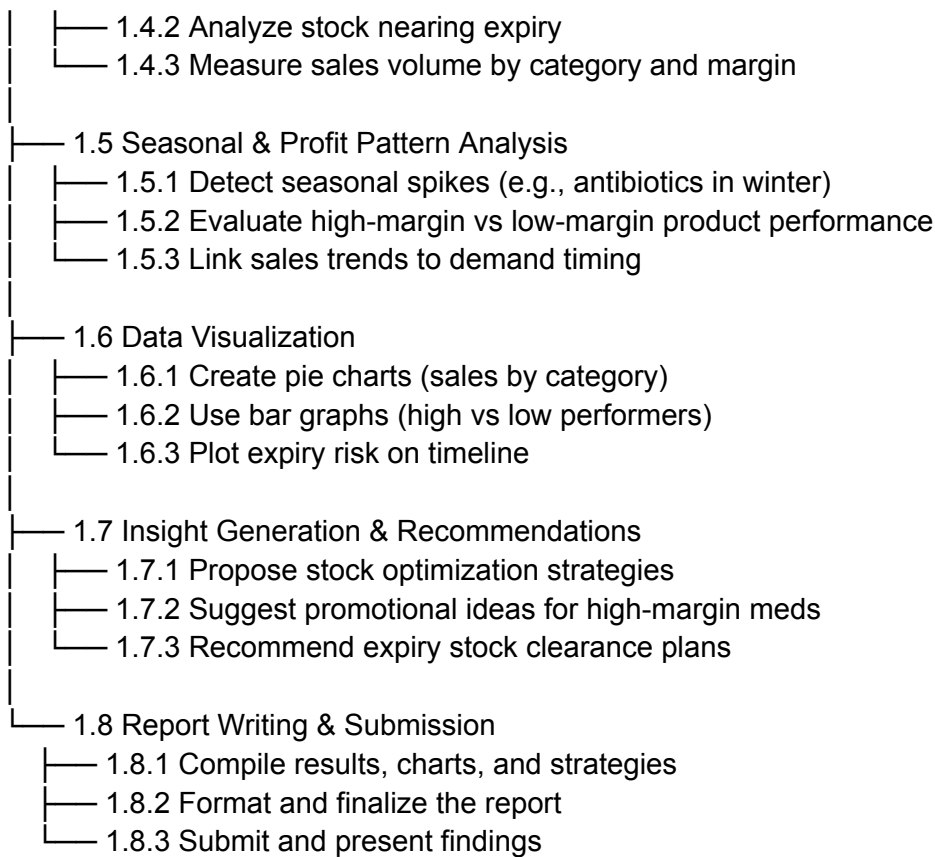
6. Expected Timeline

WBS Code	Task Name	Description
1.0	Project Initiation	Define project scope, understand medical store operations, and set objectives
1.1	Data Collection	Collect medicine sales, expiry, pricing, and quantity data from store records
1.2	Data Cleaning & Preparation	Organize data from handwritten notes; standardize “Particulars”, “Qty.”, “Date”
1.3	Sales and Expiry Analysis	Identify fast-moving vs slow-moving medicines, expiry risk, and profit margins
1.4	Seasonal & Profit Margin Trends	Analyze patterns (e.g., antibiotics in winter, Betadine sales, etc.)
1.5	Visualization of Insights	Create graphs to show inventory status, expiry risk, and margin performance
1.6	Recommendations & Optimization Plan	Suggest stock planning, discounting, supplier returns, and promotions
1.7	Final Report Compilation	Prepare a structured final report with all findings and visual elements

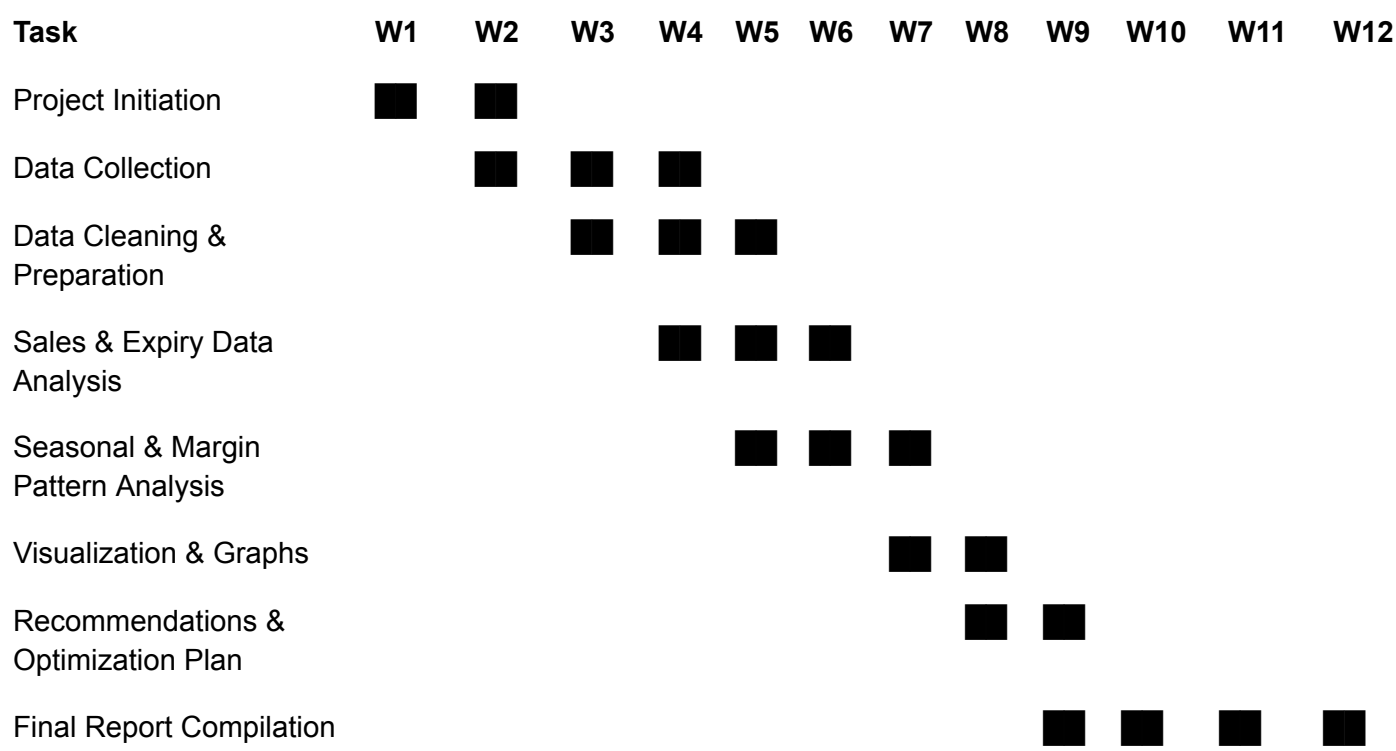
6.1 Work Breakdown Structure – Tree Format

1.0 Project: Business Data Analysis of Shri Balaji Medical Store





6.2 Gantt Chart (12 Weeks)



7. Expected Outcome

The key outcome of this Business Data Management project will be to generate valuable insights that improve the operational efficiency of **Shri Balaji Medical Store**. Through systematic collection and analysis of sales, inventory, and expiry-related data, the store will gain a clearer understanding of **high-demand medicines**, enabling **optimized stock planning** and reducing both **stockouts** and **overstocking** of low-demand items.

The project will also help uncover reasons behind **product returns**—whether due to **expiration**, **misdispensing**, or **customer dissatisfaction**—allowing the store to reduce financial losses and improve service quality. Additionally, analysis of sales trends over time will reveal **seasonal demand patterns**, such as higher antibiotic sales during winter, while customer data will provide basic **demographic insights** (e.g., age groups, common ailments).

These insights will support smarter business decisions, from **inventory control** to **targeted promotions**. Furthermore, the project will provide the groundwork for **future digital record-keeping** or **pharmacy management software adoption**, helping the store transition from manual operations to a **data-driven, efficient, and scalable model**. Ultimately, the project aims to increase **profitability**, **regulatory compliance**, and **customer satisfaction**, making the business more competitive and sustainable.